

Air Quality Dataset – Data Cleaning

1. Dataset Source

The dataset used for this project is an **Air Quality dataset** (`airquality.csv`) containing weather and environmental measurements such as temperature, pressure, humidity, wind speed, and weather condition.

The dataset was loaded into Python using the Pandas library and processed using Google Colab.

2. Dataset Size Before Preprocessing

Before preprocessing, the dataset contained:

- **Rows:** 3,500
- **Columns:** 11

Columns

1. `record_id`
2. `date`
3. `time`
4. `season`
5. `month`
6. `day_of_week`
7. `temperature`
8. `pressure`
9. `humidity`
10. `wind_speed`
11. `weather_condition`

The initial dataset dimensions were checked using Pandas `df.shape`.

3. Problems Identified

The following data-quality issues were identified:

Missing Values

Missing values were present in several columns, including:

- `date`
- `time`
- `temperature`

- pressure
- humidity
- wind_speed
- weather_condition

The preprocessing code specifically identifies non-numerical columns and removes rows containing missing values in those columns.

Outliers

Potential outliers in numerical data were identified using the **Interquartile Range (IQR) method**.

The IQR method uses:

- $Q1 = 25\text{th percentile}$
- $Q3 = 75\text{th percentile}$
- $IQR = Q3 - Q1$
- Lower Bound = $Q1 - 1.5 \times IQR$
- Upper Bound = $Q3 + 1.5 \times IQR$

Values outside these boundaries were treated as potential outliers.

4. Preprocessing Techniques Applied

The following preprocessing techniques were applied:

4.1 Handling Missing Categorical/Time Values

Non-numerical columns were identified and rows containing missing values in these columns were removed.

This was done using:

```
dropna(subset=cat_time_cols)
```

4.2 Outlier Detection

The **IQR method** was used to identify potential outliers in numerical columns.

4.3 Outlier Replacement

Detected numerical outliers were replaced with `NaN` so that they could be processed during the next step.

4.4 Interpolation

Linear interpolation was applied to numerical columns to fill missing values created during outlier removal.

The interpolation was performed in both directions to handle missing values at the beginning or end of the dataset.

4.5 Mean Imputation

If any numerical missing values remained after interpolation, they were replaced with the mean value of the respective column.

5. Dataset Size After Preprocessing

After preprocessing, the cleaned dataset contains:

- **Rows:** 2,565
- **Columns:** 11

Therefore:

Rows removed = 3,500 – 2,565 = 935 rows

The number of columns remained unchanged because the preprocessing process removed problematic rows rather than deleting columns.

6. Final Cleaned Dataset

The final cleaned dataset is:

```
airquality_cleaned.csv
```

It contains:

- **2,565 rows**
- **11 columns**
- No remaining missing values
- The original 11 features are retained

The cleaned dataset was saved as a CSV file after preprocessing. The code also prints the final dataset dimensions after cleaning.

7. Summary

Stage	Rows	Columns
Before preprocessing	3,500	11
After preprocessing	2,565	11
Rows removed	935	0

Preprocessing Summary

- Missing categorical/time values → Removed affected rows
- Numerical outliers → Detected using IQR
- Outliers → Replaced with `NaN`
- Missing numerical values → Linear interpolation
- Remaining numerical missing values → Mean imputation
- Final dataset → `airquality_cleaned.csv`

8. Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

The cleaning script also generates a correlation heatmap and a distribution plot after preprocessing.